Recovering Markov dynamics from noised sequences: a ladder of exactly solvable cases

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Abstract

The global minimiser of the denoising objective that trains a diffusion model is not the score of the data distribution: it is the score of the *empirical* distribution, a sum of point masses at the training samples smoothed by the forward noise. A model holding that score exactly would reproduce its training set and nothing else, yet diffusion models generate new data. Deciding what a trained network has actually learned requires comparing it against the score it should have learned — and for essentially every distribution of interest that target is unavailable, so a network's error cannot be separated from the difficulty of its target. The analytical literature therefore works with Gaussians and Gaussian mixtures, where the score is linear and any question about how structure is exploited has a degenerate answer.

We study a family where the target is computable and non-trivial: a first-order Markov chain on $\mathbb{R}^L$ with continuous, non-Gaussian transitions, variance-matched so that innovation families differ only beyond second moments. Coordinatewise Ornstein–Uhlenbeck noising multiplies the prior by *unary* factors, adding no edges among the latent variables, so the posterior factor graph is still the prior's chain — a tree, on which sum-product is exact. The exact score is therefore available for a correlated, non-Gaussian model.

We then drop the assumption that the chain's parameters are known and learn them from noised sequences alone. Fisher's identity supplies the exact gradient of the marginal log-likelihood from a single sum-product pass, with no differentiation through the recursion, so both gradient ascent and expectation–maximisation are available; for Gaussian innovations the maximisation step is closed form and the two routes reach the same optimum. We prove that single-Gaussian message closure computes *exactly* the LMMSE estimator of the covariance-matched Gaussian model, so the gap between it and full sum-product is precisely the price of discarding everything beyond second moments. Estimating a transition *density* rather than its parameters exposes a convergence-rate asymmetry, which we trace to the innovation mixture's own latent label rather than to the channel.

Against networks trained by denoising score matching on identical data, the structured estimator attains between 8 and 14 times lower pointwise denoising error across seven doublings of the training set, and the network at the largest budget does not reach what the estimator attains at the smallest.

The last rung leaves the chain. On a rotating ring the rotation acts as a gauge, and every single-frame marginal is provably independent of it, so a per-frame model carries exactly zero information about the dynamics. EM on the joint recovers the

rotation to $0.15°$ at low noise; EM on the marginals cannot move off its initialisation, and does not — 0.0000 in total variation across all $1{,}344$ cells, which is what a theorem predicting exactly no information requires.

1 Introduction

Generative diffusion produces data $\mathbf{a} \in \mathbb{R}^L$ from a distribution P_0 known only through samples, by noising and then reversing [1–3]. Reversal is driven by the score $S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x})$, which is learned by regression against the posterior mean of the clean signal.

The puzzle that motivates this. A network never sees P_0. It sees M samples, which is to say the empirical measure $\hat{P}_0 = M^{-1} \sum_\mu \delta(\cdot - \mathbf{a}^\mu)$, and the global minimiser of the denoising objective is the score of *that* — the empirical score, a sum of point masses smoothed by the forward Gaussian. A model attaining it exactly would have a reverse process whose fixed points are the training samples, and would generate nothing else. Diffusion models nonetheless generate new data, and understanding when they memorise and when they generalise is an active question in the statistical mechanics of these models [4, 5].

Measuring this is the difficulty. Deciding what a trained network has learned means comparing it against the score it *should* have learned, and for realistic data that object is unavailable — so a large error may indicate a poor model or a hard target, and nothing in the measurement distinguishes them. Distributions with tractable population scores are scarce, which is why the analytical literature leans on Gaussians and Gaussian mixtures; those are poor probes of *structure*, since the score is linear and any question about exploiting correlation has a degenerate answer.

Why a Markov chain. Two reasons. Diffusion is applied to sequential data — video, audio, time series — so a model of correlation along a sequence is the relevant caricature. And studying how structure is exploited requires data that has structure, of a kind one can dial. A first-order Markov chain with continuous non-Gaussian transitions supplies both, and the variance-matching of §3 makes the innovation family a controlled knob: Gaussian, Laplace, Student, uniform and bimodal chains share a covariance exactly and differ only beyond it.

Why it is tractable. Coordinatewise noising contributes one *unary* factor per site. Unary factors reweight node potentials without creating edges among the latent variables, so conditioning leaves the factor graph a chain — a tree, on which sum-product returns exact marginals. The exact score is therefore computable for a correlated, non-Gaussian model, and the only obstruction is that the messages are functions on $\mathbb{R}$ and must be represented. The Gaussian case, where the answer is known in closed form, is what validates that representation before it is used where the answer is not known.

The question this makes askable. If the data is Markov, is it *easier to learn*? The claim sounds trivial; the content is in saying why, and in measuring it. Sum-product reaches the exact score through very simple operations, and it is Markovianity that permits this. So we drop the assumption that the chain's parameters are known: keep the recursion, treat the transition kernel as unknown, start from a random initialisation, and ascend the likelihood in the parameters on a finite sample. Fisher's identity makes the exact gradient available from one sum-product pass. The resulting estimator is then compared against networks trained by denoising score matching on identical data.

One question, asked five times. Rather than survey what the model can do, we ask a single question at four levels of difficulty and derive the answer at each before measuring it. The question is: *given only sequences seen through the noising channel, what can be recovered?* The rungs are

	what is unknown	what is derived	§
0	nothing — the kernel is given	the posterior mean, in closed form	4
1	two numbers (ρ, q)	exact gradient; closed-form maximisation	5
2	a transition *density*	a second latent; a rate asymmetry	6
3	dynamics invisible to marginals	a blindness theorem	7

Each rung adds exactly one difficulty to the one below it, and each experiment tests a statement derived beforehand. Nothing appears below that is not on a rung.

Contributions.

- An exact denoising reference for a continuous, non-Gaussian, correlated model, with truncation and quadrature error separated and measured, validated against a closed form and against brute-force enumeration (§4).
- An identification of what single-Gaussian message closure computes on this model: exactly the LMMSE estimator of the covariance-matched Gaussian model (Proposition 1), proved directly rather than argued asymptotically.
- Estimation of the kernel from noised sequences alone, by gradient ascent or EM, with the exact gradient from one sum-product pass and no differentiation through the recursion (§5–§6).
- A measurement of what the Markov structure is worth against trained denoisers (§8).
- A model in which the dynamics is provably invisible to every single-frame marginal and still recoverable from the joint (Theorem 1, §7).

2 Related work

Diffusion and score matching. The construction is due to Sohl-Dickstein et al. [1], developed by Ho et al. [2] and given its continuous-time form by Song et al. [3]; training targets the score by denoising score matching [6, 7], and that the regression target is a posterior mean is Tweedie's formula [8–10].

Statistical mechanics, and the memorisation question. The high-dimensional behaviour of these models is analysed as a phase-transition problem [4, 5, 11], identifying distinct regimes along the schedule and the point past which the reverse process commits to a mode. The puzzle above is attacked within it from several directions: through training dynamics, where the time to memorise grows with sample size while the time to generalise does not [12, 13]; through the learned representation, geometry-adaptive rather than sample-specific [14]; through the spectrum of the empirical score [15]; and in solvable settings [16–18]. The window is not clean: Garnier-Brun et al. [19] exhibits a phase where test loss still falls while samples drift anomalously close to the training set. Every one of these measures a network against a target unavailable in closed form — which is what the model below supplies.

Tractable structured data, and architectures that implement inference. Hierarchical generative models supply tractable non-Gaussian families with a phase structure [20], and the same construction can be run backwards to read latent structure off real data [21] or to date the emergence of compositional generalisation during training [22]. Two results bear directly on what we leave open: Mei [23] reads U-Nets on such models as implementing belief propagation, and Garnier-Brun et al. [24] shows transformers learning the corresponding inference on hierarchically filtered data. Both concern *tree*-structured data, where the natural depth is the tree's. A chain is a degenerate tree — depth equal to its length, width two — so the architectural implication differs, which is what §8 sets up rather than settles.

Graphical models and inference. Sum-product and its exactness on trees are standard [25–28]; on a chain the recursions are forward–backward [29, 30], reducing to Kalman and RTS in the Gaussian case [31, 32]. The estimation machinery is EM [33], its conditional-maximisation variants [34, 35], and Fisher's identity [36–38].

3 Setup

Notation. L is the chain length, which is also the ambient dimension, and M is the number of observed sequences; the diffusion literature's n and d translate as $n \to M$, $d \to L$. Vectors are bold lowercase and matrices bold uppercase, so $\mathbf{a} \in \mathbb{R}^L$ is a whole clean chain and a_i its i-th site. K is always the transition kernel and N_{g} always the number of quadrature points (Appendix A).

Forward process. With η standardised Gaussian white noise, $\langle \eta(t)\eta(t')\rangle = \delta(t-t')$, the forward Ornstein–Uhlenbeck process and its solution are

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} \;=\; -\mathbf{x} + \sqrt{2}\,\eta(t), \qquad \mathbf{x}(t) \;=\; e^{-t}\mathbf{a} + \sqrt{\Delta_t}\,\mathbf{z}, \qquad \Delta_t = 1 - e^{-2t}, \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})\,. \tag{1}$$

The reverse-time convention is fixed once in Appendix A and used without variation.

Prior. P_0 factorises as

$$P_0(\mathbf{a}) \;=\; \mu(a_1)\prod_{i=2}^{L} K(a_i \mid a_{i-1}), \qquad K(a' \mid a) \;=\; \varphi(a' - \rho a), \tag{2}$$

with φ an innovation density of mean zero and variance q, and $|\rho| < 1$. We draw $a_1 \sim \mathcal{N}(0,1)$ and set $q = 1 - \rho^2$, which makes $\mathrm{Var}(a_i) = 1$ at every site and $\mathrm{Cov}(a_i, a_j) = \rho^{|i-j|}$ exactly, *whatever the shape of* φ. Variance matching is what makes the family a controlled probe: Gaussian, Laplace, Student, uniform and bimodal innovations share the same covariance and differ only beyond second moments. The chain is covariance-stationary but not strictly stationary; Appendix E states the difference, and shows why it collapses in the Gaussian case.

The score is a posterior mean. From (1), P_t is a Gaussian convolution of P_0. Differentiating under the integral and dividing by $P_t(\mathbf{x})$ — the Gaussian factor over $P_t(\mathbf{x})$ is exactly the posterior density $P(\mathbf{a} \mid \mathbf{x}, t)$ — gives

$$S(\mathbf{x}, t) \;=\; -\frac{\mathbf{x} - e^{-t}\,\langle \mathbf{a}\rangle_{x,t}}{\Delta_t}, \qquad \langle \mathbf{a}\rangle_{x,t} = \int \mathrm{d}\mathbf{a}\;\mathbf{a}\,P(\mathbf{a} \mid \mathbf{x}, t). \tag{3}$$

Equation (3) is a change of variables, not an approximation: computing the score and computing the posterior mean are the same problem. The identity is classical in the empirical-Bayes literature [8–10] and underlies denoising score matching [6, 7]. For squared loss $\langle \mathbf{a}\rangle_{x,t}$ minimises the conditional risk (Appendix B), so it is the reference every estimator below is scored against.

Conditioning preserves the chain. Conditioning (2) on $\mathbf{x}$ gives

$$P(\mathbf{a} \mid \mathbf{x}, t) \;\propto\; \mu(a_1)\prod_{i=2}^{L} K(a_i \mid a_{i-1})\prod_{i=1}^{L} \ell_t(x_i \mid a_i), \qquad \ell_t(x_i \mid a_i) \propto \exp\!\left[-\frac{(x_i - e^{-t}a_i)^2}{2\Delta_t}\right]. \tag{4}$$

The likelihood factors are *unary*: the forward noise is independent across coordinates, so each observation touches one latent variable. They reweight node potentials and add no edges, so the posterior factor graph is the prior's chain (drawn in Appendix D).

A *tree* is a connected acyclic graph — there is exactly one path between any two nodes. A chain is the simplest tree, the one whose nodes have degree at most two. On a tree, sum-product returns the exact marginals [25–27]: with

$$\begin{aligned}
\overrightarrow{m}_i(a_i) &\propto \int \mathrm{d}a_{i-1}\; K(a_i \mid a_{i-1})\,\ell_t(x_{i-1} \mid a_{i-1})\,\overrightarrow{m}_{i-1}(a_{i-1}),\\
\overleftarrow{m}_i(a_i) &\propto \int \mathrm{d}a_{i+1}\; K(a_{i+1} \mid a_i)\,\ell_t(x_{i+1} \mid a_{i+1})\,\overleftarrow{m}_{i+1}(a_{i+1}),
\end{aligned} \tag{5}$$

initialised by $\overrightarrow{m}_1 = \mu$ and $\overleftarrow{m}_L \equiv 1$, the single-site marginal is $P(a_i \mid \mathbf{x}, t) \propto \overrightarrow{m}_i(a_i)\ell_t(x_i \mid a_i)\overleftarrow{m}_i(a_i)$ and the pairwise marginal on edge $(i-1, i)$ is $\propto \overrightarrow{m}_{i-1}\ell_t(x_{i-1} \mid a_{i-1})K(a_i \mid a_{i-1})\ell_t(x_i \mid a_i)\overleftarrow{m}_i$. The local likelihood is absorbed *before* propagation, so $\ell_t(x_i \mid a_i)$ appears exactly once in each marginal. These are the forward–backward recursions of hidden Markov models [29, 30] written for a continuous state; when everything is Gaussian they reduce to the Kalman filter and RTS smoother [31, 32].

Two cautions, both expanded in Appendix C. Sum-product is exact on the tree, but a message is a *function* on $\mathbb{R}$ — forced by the state space, not chosen — so an implementation must represent one, and the truncated grid we use makes the recursion exact only for a finite-state approximation whose error §4 measures. And Monte Carlo enters this note in exactly one place, when a statistic is computed from generated samples; every other number comes from deterministic quadrature and carries discretisation error but no sampling error.

Cost. Direct summation over the N_{g}^{L} joint configurations costs $O(N_{\mathrm{g}}^{L-1})$ — about 10^{80} terms at $N_{\mathrm{g}} = 401$, $L = 32$. The recursion replaces it with $2(L-1)$ matrix–vector products: $O(LN_{\mathrm{g}}^{2})$ time per sequence and $O(LN_{\mathrm{g}})$ storage for the messages, plus one shared $O(N_{\mathrm{g}}^{2})$ transition matrix. Exponential in the chain length becomes linear in it and quadratic in the representation size (Appendix D).

4 Rung 0 — the kernel is known

With K given, (5) returns the posterior mean and hence, through (3), the score. Two things must be established before anything is built on it: that the discretisation is controlled, and what the standard Gaussian approximation actually computes.

The discretisation, controlled on two axes. Two error sources are separated by construction: *truncation*, mass outside $[-A, A]$ discarded rather than transported, controlled by A and essentially independent of N_{g}; and *quadrature*, second order in h, and no better than second order for the Laplace family, whose innovation has a discontinuous derivative at the origin. Each is measured with a diagnostic that isolates it, since a resolution sweep at fixed A cannot rule out truncation and a domain sweep at fixed N_{g} cannot rule out quadrature (Figure 4, Appendix G). The working configuration $N_{\mathrm{g}} = 401$, $A = 8$ is read off those curves, and puts both diagnostics far below every effect reported here.

Validation where the answer is known. Take $\varphi = \mathcal{N}(0, q)$. Then $\mathbf{a}$ is jointly Gaussian with $(\boldsymbol{\Sigma}_0)_{ij} = \rho^{|i-j|}$, and so is $\mathbf{x}$, with $\boldsymbol{\Sigma}_t = e^{-2t}\boldsymbol{\Sigma}_0 + \Delta_t\mathbf{I}$, giving

$$\langle \mathbf{a} \rangle_{x,t} = e^{-t}\boldsymbol{\Sigma}_0\boldsymbol{\Sigma}_t^{-1}\mathbf{x}, \qquad S(\mathbf{x}, t) = -\boldsymbol{\Sigma}_t^{-1}\mathbf{x}. \tag{6}$$

The clean precision $\boldsymbol{\Sigma}_0^{-1}$ is tridiagonal, and the posterior precision $(e^{-2t}/\Delta_t)\mathbf{I} + \boldsymbol{\Sigma}_0^{-1}$ stays tridiagonal at every t: being Markov is a statement about the precision, not the covariance, which is dense. At $N_{\mathrm{g}} = 401$, $A = 8$ the discretised recursion reproduces (6) to 9.2×10^{-15} relative, and agrees with brute-force enumeration of the discretised posterior — summing over all N_{g}^{L} configurations on a short chain, sharing no code with the message passing — to 1.6×10^{-14} on posterior means and 1.8×10^{-15} on the log-evidence. Neither check involves non-Gaussian machinery: the pipeline is verified where the answer is known before being used where it is not.

What the standard Gaussian approximation computes. A common approximation propagates only the first two moments of each message. Under $a_i = \rho a_{i-1} + \varepsilon_i$, a message with mean m and variance v transforms as

$$\text{forward:} \quad m \mapsto \rho m, \;\; v \mapsto \rho^2 v + q; \qquad \text{backward:} \quad m \mapsto m/\rho, \;\; v \mapsto (v+q)/\rho^2, \tag{7}$$

the backward map being the forward one inverted because that recursion integrates over the output variable (it requires $\rho \neq 0$). The propagation uses only the mean and variance of φ.

Proposition 1. *For the chain (2) with linear transitions, Gaussian unary likelihoods and $\mathbb{E}[\mathbf{a}] = \mathbf{0}$, moment-projected sum-product returns the LMMSE estimator of the covariance-matched Gaussian model,*

$$\widehat{\mathbf{a}}_{\mathrm{LMMSE}} = \mathrm{Cov}(\mathbf{a}, \mathbf{x})\ \mathrm{Cov}(\mathbf{x})^{-1}\mathbf{x} = e^{-t}\boldsymbol{\Sigma}_0\left(e^{-2t}\boldsymbol{\Sigma}_0 + \Delta_t\mathbf{I}\right)^{-1}\mathbf{x}, \tag{8}$$

for every innovation law with mean zero and variance q.

Proof sketch. Both maps in (7) read φ only through its mean and variance, so the recursion returns the same function of $\mathbf{x}$ for every innovation with moments $(0, q)$ — in particular the same as on the covariance-matched Gaussian chain, where the moment projection is the identity and the recursion returns the posterior mean, which for jointly Gaussian zero-mean $(\mathbf{a}, \mathbf{x})$ is linear and hence the best linear estimator. Appendix F. □

We call (8) the *Gaussian second-order baseline*. On this model, message closure, moment projection and outright replacement of the prior by its covariance-matched Gaussian all coincide; they separate for richer message families. We do not call the method AMP: a degree-two chain supplies no large-degree central-limit justification for that terminology, and Proposition 1 is proved directly rather than argued asymptotically. The zero-mean hypothesis is load-bearing — with $\mathbb{E}[a] = 1.3$ the estimator (8) departs from the true LMMSE by 0.65 in maximum absolute value, because the general form carries $\mathbb{E}[\mathbf{a}] + \mathrm{Cov}(\mathbf{a}, \mathbf{x})\,\mathrm{Cov}(\mathbf{x})^{-1}(\mathbf{x} - \mathbb{E}[\mathbf{x}])$.

Proposition 1 is what makes the gap between (8) and grid BP interpretable: it is not an artefact of a message representation, it is exactly the price of replacing the prior by its covariance twin. That gap locates where non-Gaussianity matters (Figure 1), and it shrinks as t grows because large t Gaussianises the posterior — the shape of φ survives only where the likelihood is weak.

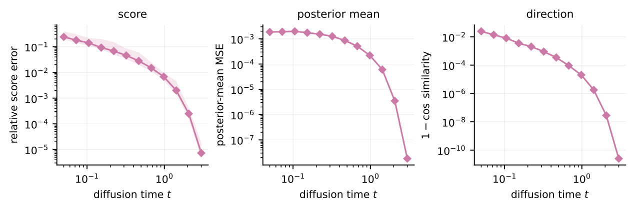

Figure 1: The cost of the Gaussian second-order baseline, against grid BP under the true kernel. Laplace chain, $\rho = 0.85$, $L = 32$, median over seeds with the 10–90 percentile band. By Proposition 1 this gap is exactly the price of replacing the prior by its covariance-matched Gaussian. It vanishes as t grows, because noising Gaussianises the posterior.

5 Rung 1 — recovering two numbers

Now K is unknown. Take the simplest case first: Gaussian innovations, so the kernel is $K_\theta(a' \mid a) = \mathcal{N}(a' - \rho a, q)$ and $\theta = (\rho, q)$ is two numbers. We observe only noised sequences $\mathbf{x}^\mu$, $\mu = 1, \ldots, M$, and write $\mathcal{L}(\theta) = \sum_\mu \log P_{t,\theta}(\mathbf{x}^\mu)$ for the marginal log-likelihood.

The gradient is free. The obvious route is gradient ascent on $\mathcal{L}$. That looks expensive — $\mathcal{L}$ is defined through the recursion, so differentiating it appears to require backpropagating through $2(L-1)$ message updates. It does not. Fisher's identity [36–38] gives

Proposition 2 (Exact gradient from one sum-product pass)**.** *With* $\boldsymbol{\Xi}$ *the expected transition mass between grid points,*

$$\boldsymbol{\Xi}_{kl} \;=\; \sum_{\mu=1}^{M} \sum_{i=2}^{L} P(a_{i-1} = u_k,\; a_i = u_l \mid \mathbf{x}^\mu)\,, \tag{9}$$

the gradient of the marginal *log-likelihood is*

$$\nabla_\theta \mathcal{L}(\theta) = \sum_\mu \mathbb{E}_\theta[\nabla_\theta \log P_\theta(\mathbf{a}, \mathbf{x}^\mu) \,|\, \mathbf{x}^\mu] = \big\langle \boldsymbol{\Xi}(\theta),\; \nabla_\theta \log K_\theta \big\rangle. \tag{10}$$

Sum-product is not a computation graph to backpropagate through; it supplies the conditional expectation, and one forward–backward pass yields the exact gradient. We verify (10) against finite differences of the discretised evidence to $\sim 10^{-9}$.

$\boldsymbol{\Xi}$ is the continuum analogue of the expected transition counts of Baum–Welch [29], and it is *sufficient*: the complete-data log-likelihood is a sum over edges, so the θ-dependent part of the auxiliary function $\mathcal{Q}(\theta \mid \theta^{(r)})$ depends on the data only through $\boldsymbol{\Xi}$. Sufficiency requires a homogeneous kernel shared across sites and sequences, a fixed grid, an initial law excluded from θ, and θ-free likelihood factors — the last being what licenses observations at several noise levels to add into one $\boldsymbol{\Xi}$. Given those, maximisation never revisits the data and costs $O(PN_\mathrm{g}^2)$ for P parameters, independently of M. Forming $\boldsymbol{\Xi}$ costs $O(MLN_\mathrm{g}^2)$ per iteration, so "one matrix" is a statement about sufficiency, not about inference being cheap.

Two routes to the same place. Equation (10) licenses gradient ascent; EM instead increases $\mathcal{Q}$. For the Gaussian kernel the maximisation step is closed form — profiling $\mathcal{Q}$ gives the belief-weighted Yule–Walker equations

$$\rho \;=\; S_{01}/S_{00}, \qquad q \;=\; (S_{11} - \rho\, S_{01})/W, \tag{11}$$

with $S_{ab} = \sum_{kl} \Xi_{kl}\, u_k^a u_l^b$ and $W = \sum_{kl} \Xi_{kl}$ — so this rung has *exact* EM, not a generalised variant. The derivation is Appendix H.

Both routes are implemented and neither dominates. On this smooth kernel gradient ascent reaches EM's optimum to 1.6×10^{-9} nats, but it needs a tuned step size and larger steps diverge; EM needs no step size and its ascent is monotone by construction [33, 34]. That is the whole content of the comparison, and it is why every rung above uses EM. The one place gradient ascent wins is a kernel whose $\mathcal{Q}$-maximiser is a lattice artefact of the discretisation, which is a statement about the grid rather than about the algorithms (Appendix H).

Identifiability, at this rung, is a theorem. (11) inverts the map $(\rho, q) \mapsto (S_{01}/S_{00}, \dots)$ explicitly, and Gaussian convolution at finite t is injective on distributions — the characteristic function acquires a factor $e^{-\Delta_t \|\xi\|^2/2}$, which has no zeros, and $e^{-t} > 0$ — so P_t determines P_0 and P_0 determines (ρ, q). Nothing is assumed here. That changes at Rung 2, and the change is worth marking rather than burying: see §6.

What is measured. Recovery of (ρ, q) from initialisations away from the truth, and the rate at which the error falls with M. Theory predicts $M^{-1/2}$, the ordinary parametric rate, since this is maximum likelihood in a correctly specified two-parameter model. Over $M = 64$ to 1024 at sixteen replicates, the innovation-variance error falls from 2.7×10^{-2} to 6.4×10^{-3}, a fitted log–log slope of -0.52 against the predicted -0.50 ($r = -0.99$, monotone). The ρ error falls monotonically over the same range but more slowly, at -0.27; we report the discrepancy rather than explain it, since at these budgets ρ is already accurate to 5×10^{-3} and we have not established that the residual is sampling error rather than a floor.

6 Rung 2 — recovering a density

Free the innovation shape. Let

$$K_\theta(a' \mid a) \;=\; \sum_{c=1}^{C} \pi_c\, \mathcal{N}\big(a' - \rho a - \nu_c, s_c^2\big)\,, \tag{12}$$

in which **both** the autoregressive coefficient and the innovation density are free. At $C = 8$ this is ρ, seven weights after the simplex constraint, eight locations and eight scales. Two things change at once, and they are worth separating.

A second latent variable. At Rung 1 the complete-data problem — knowing $\mathbf{a}$, fit K — was a two-parameter regression with a closed-form solution. Here, knowing $\mathbf{a}$ is not enough: the complete-data problem is itself a mixture fit, carrying the component label as a second latent variable, so the maximisation step has no closed form. We run four inner conditional-maximisation sweeps, alternating the (π, ν, s^2) and ρ blocks, each closed form given the other. The algorithm is therefore **generalised EM** of ECM type [33, 35]: every step increases $\mathcal{Q}$ without maximising it, which preserves monotone ascent [34] but not the exact-maximiser property. We do not call it exact EM.

Misspecification. When the truth is Laplace and the family is the mixture (12), what is recovered is a *density*, not a parameter vector, and the mixture is a flexible representation of it rather than the truth in disguise. Identifiability weakens correspondingly: finite mixtures are identifiable up to label permutation on the interior of the parameter set — all $\pi_c > 0$, the pairs (ν_c, s_c) distinct [39, 40] — which is exactly where an over-specified C need not stay. The condition is monitored (minimum component mass, minimum width in grid spacings) rather than assumed. This is weaker than Rung 1's theorem and stronger than what we can say at Rung 3.

Why the fourth cumulant, and not some other summary. A density needs a scalar summary to be tracked, and the choice is forced rather than free. Every innovation family used here is symmetric, so all odd cumulants vanish: $\kappa_3 = 0$ identically. The variance is matched by construction, $q = 1 - \rho^2$, so κ_2 is fixed across families and carries no discriminating information at all. The first cumulant that *can* differ between a Laplace innovation and its covariance twin is therefore the fourth. Excess kurtosis is not a preference; at matched variance and under symmetry it is the leading coordinate in which non-Gaussianity can appear.

The coordinates do not converge at the same rate. This is the substantive finding of the rung, and it is a statement about the optimiser rather than about information.

Remark 1 (Two different slownesses, kept apart)**.** *On* clean *chains, with no missing information whatsoever, ρ and q are converged after a single maximisation step — with clean data Ξ is the empirical pair distribution and does not depend on θ at all — while the innovation shape still takes tens of iterations. That slowness belongs to the mixture's inner latent label, the extra difficulty introduced above. It is not caused by the channel.*

Through the channel both coordinates slow down further, and that *is the channel's doing: it is the missing-information mechanism of Dempster et al. [33] appearing exactly where the theory puts it, in the convergence rate rather than in the attainable accuracy.*

The two effects are separable and we separate them, because conflating them produces a false conclusion — an apparent information *penalty that is really a rate at a fixed iteration budget. Both arms recover the truth when run to convergence.*

The practical consequence is a stopping rule. The autoregressive coefficient settles roughly an order of magnitude sooner than the shape, and a ρ-trace looks textbook-converged over a kernel that is nowhere near converged. Any stopping rule for this model must therefore be read off a *shape* statistic; Appendix K states the one used, and it is the same rule at every rung. The fitted density itself is Figure 5 in Appendix I; the settling iterations, and the clean-against-channel comparison that separates the two slownesses, are Appendix I.

7 Rung 3 — dynamics the marginals cannot see

Every rung so far lives on the chain, where what is estimated is visible in low-order statistics of the data. The last rung asks the same question where it is not.

The model. A point starts on a ring of radius $\approx$1 at a uniformly random angle and takes a rotating random walk in the plane:

$$z_0 = r_0(\cos\theta_0, \sin\theta_0), \quad r_0 \sim \rho\, e^{-(\rho-1)^2/2\lambda}, \;\; \theta_0 \sim U[0, 2\pi), \qquad z_{u+1} = R_\psi z_u + \sigma\eta_u. \quad (13)$$

One sample is a *whole* trajectory of T frames, $\mathbf{a} \in \mathbb{R}^{2T}$, and the channel (1) hits all of it at once. The rotation angle ψ is drawn per trajectory from an unknown $p(\psi)$, which is what we estimate.

The rotation is a gauge. $U_\psi = \mathrm{blockdiag}(R_{-u\psi})$ is orthogonal, so it commutes with the isotropic channel and maps the model onto $\psi = 0$, a plain planar random walk started on the ring: $P_t(\mathbf{x} \mid \psi) = P_t^0(U_\psi \mathbf{x})$. The ψ-conditional likelihood is therefore closed form at any T — rotate the data back and evaluate one quadratic form plus a single one-dimensional Bessel integral. No network, no Monte Carlo, no curse of dimension (Appendix J).

Theorem 1 (Marginal blindness)**.** *For the model (13) with uniform initial phase, every single-frame marginal is independent of ψ, at every noise level. Consequently a per-frame model carries exactly zero information about the rotation.*

Proof. $z_u = R^u z_0 + \sigma \sum_{k\le u} R^{u-k}\eta_k$. Each $R^{u-k}\eta_k \overset{d}{=} \eta_k$ by isotropy of the Gaussian, and $R^u z_0 \overset{d}{=} z_0$ because θ_0 is uniform on the circle. Hence $z_u \overset{d}{=} z_0 + \sigma\sqrt{u}\,\xi$, in which ψ does not appear. Noising acts coordinatewise and preserves this. Full statement in Appendix J. $\square$

This is a sharper object than the usual joint-versus-marginal comparison. The per-frame marginal is not merely *less* informative about the dynamics — it is not approximately blind, or blind at small t — it carries *zero* information, exactly, at every noise level. And the marginal is not degenerate: it is a visible, non-Gaussian ring. Radii are not evidence.

EM over $p(\psi)$, and its own control. The estimator is EM for a mixture over the rotation angle on a grid of angles: the expectation step forms responsibilities $r_\mu(\psi) \propto p(\psi)P_t(\mathbf{x}^\mu \mid \psi)$ from the closed form above, and the maximisation step sets $p(\psi) \leftarrow M^{-1}\sum_\mu r_\mu(\psi)$. As at Rungs 1–3 the data-dependent quantity is computed once: the ψ-conditional log-likelihood does not depend on $p(\psi)$, so the $M \times |\psi|$ matrix is formed once and every iteration is a reweighting of it — the same sufficiency that makes Ξ a one-shot statistic on the chain.

Theorem 1 supplies the control, and it is not a strawman baseline. Score the same trajectories with the per-frame marginal instead of the joint: the likelihood is then constant in ψ, so the responsibilities equal $p(\psi)$ exactly, the maximisation step is the identity, and $p(\psi)$ can never leave its uniform initialisation. The theorem predicts an exact zero, and the run either delivers exactly zero or the implementation is wrong.

It delivers exactly zero. Over $1{,}344$ cells — sixteen seeds, seven budgets, twelve noise levels — the blind arm's estimate moves 0.0000 in total variation from its uniform initialisation, and its profile likelihood in ψ has range 0.00×10^0 nats across the 480 cells of the single-rotation variant. Not a small number: a machine-precision zero, everywhere, which is what a theorem predicting exactly no information requires.

The joint arm recovers the rotation, and degrades with noise as one would expect of a quantity being destroyed by a channel. Estimating a single rotation angle it is accurate to $2.1 \pm 0.7°$ pooled over the whole schedule; recovering the two-species law $p(\psi)$, the median error in ω runs

t	0.05	0.32	0.68	0.98	1.43	3.00
median $\lvert\hat{\omega} - \omega\rvert$	$0.15°$	$0.24°$	$0.90°$	$2.86°$	$7.09°$	$58.8°$

against a truth of $30°$ and a null of $90°$. Information about the rotation survives the channel until $t \approx 1$ and is gone by $t = 3$, where the estimate has returned to the null.

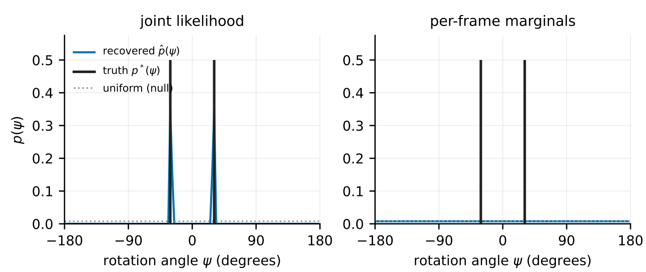

Figure 2: Rung 4. Left: the recovered $p(\psi)$ against the two-species truth $\frac{1}{2}\delta_{+\omega} + \frac{1}{2}\delta_{-\omega}$; EM on the joint finds both species. Centre: the same run scored with per-frame marginals, which by Theorem 1 cannot move off uniform, and does not. Both panels share a y-axis, which is the only honest way to draw the comparison.

8 What the structure buys

The rungs above establish what is recoverable. This section asks what the structural assumption is worth against a generic function approximator, and is deliberately the shortest in the note.

Remark 2 (Information asymmetry)**.** *The estimator is given the Markov factorisation, the kernel's linear-autoregressive form, and homogeneity across sites; the networks are given none of these. What follows measures the value of that structure, not anything about neural denoisers in general.*

Neural baselines are trained by denoising score matching in both standard parameterisations (predicting the noise, and predicting the clean signal), with a fully connected network and a weight-shared one-dimensional convolution. Both are scored against grid BP under the true kernel on held-out sequences.

Table 1 gives a ratio between 8 and 14 across seven doublings of the data. The scoped statement it supports is: *under the correctly specified Markov model and at the budgets tested, the structured estimator is more sample-efficient in pointwise denoising than the neural baselines we trained.* The network at $M = 2048$ has not reached the error the estimator attains at $M = 32$; we observe no crossing and therefore quote no data-equivalence factor.

Table 1: Relative denoising error against grid BP under the true kernel, averaged over the noise schedule; sixteen seeds $\pm$ one standard error. The network's parameterisation is chosen on a disjoint validation bundle, not the test set, and agrees with the test-set optimum in 99.4% of the $1{,}344$ cells. No crossing occurs in the range, so no data-equivalence factor is quoted. **[PENDING: EM budget: these ran at 120, the configured value is 400 (array 628943)]**

sequences M	network	EM–BP (12 free)	ratio
32	0.6613 ± 0.0026	$\mathbf{0.0703 \pm 0.0051}$	9.4
64	0.5896 ± 0.0029	$\mathbf{0.0541 \pm 0.0035}$	10.9
128	0.4942 ± 0.0015	$\mathbf{0.0361 \pm 0.0024}$	13.7
256	0.3473 ± 0.0012	$\mathbf{0.0259 \pm 0.0014}$	13.4
512	0.2359 ± 0.0010	$\mathbf{0.0233 \pm 0.0010}$	10.1
1024	0.1685 ± 0.0006	$\mathbf{0.0185 \pm 0.0011}$	9.1
2048	0.1364 ± 0.0007	$\mathbf{0.0172 \pm 0.0008}$	7.9

Protocol, in one line. Every number here comes from a single frozen configuration, defined in one file and imported by every experiment so that no run can silently diverge from it (Appendix K). The one part of it that is load-bearing rather than bookkeeping: EM stops when the fitted innovation excess kurtosis stops moving, $|\Delta\kappa_4| < 10^{-3}$, never when ρ does — by Remark 1 a settled ρ certifies a converged-looking trace over an unconverged kernel. The rule is the same at every rung.

9 Limitations

The model assumes exact first-order Markov structure, and under a non-Markov prior the estimator is misspecified in a way more data does not repair. How much that costs depends sharply on which assumption breaks: a shared global latent contaminates the prior with a rank-one term that a chain absorbs, while weak long-range precision coupling is not low-rank and cannot be represented. Both have an exact reference and are measured in the compendium; neither is claimed here.

Further scope conditions — the fixed initial law, the assumed linear kernel form, covariance-stationarity, identifiability being a theorem at Rung 1, conditional at Rung 2 and assumed at Rung 3, the second-order discretisation, the $O(LN_g^2)$ inference cost, and the composite likelihood whose effective sample size is the number of clean chains — are set out in Appendix L.

The open question is architectural, and we set it up rather than answer it: sum-product on a chain suggests a network narrow but as deep as twice the chain, unlike the tree-structured case where depth is the depth of the tree [23, 24]. Which architectures exploit Markovianity, and at what depth, is the natural continuation.

10 Conclusion

Asking one estimation question at four levels of difficulty makes it possible to say, at each, what is derived and what is measured. Moment closure is exactly an LMMSE estimator, so the gap to sum-product is the price of second-order truncation and nothing else. Fisher's identity supplies the exact gradient from one sum-product pass, which makes the kernel learnable from noised data with maximisation in closed form. Freeing the innovation density adds a latent variable and exposes a convergence-rate asymmetry belonging to the mixture rather than the channel — reading that as an information penalty is the trap this model sets, and we fell into it before separating the two. The structure is worth an order of magnitude in sample efficiency against trained denoisers. And a rotating ring gives a case where the dynamics is provably invisible to every single-frame marginal and still recoverable from the joint, with the theorem furnishing its own control.

References

[1] Jascha Sohl-Dickstein, Eric A. Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In *International Conference on Machine Learning*, 2015.

[2] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, 2020.

[3] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021.

[4] Giulio Biroli and Marc Mézard. Generative diffusion in very large dimensions. *Journal of Statistical Mechanics: Theory and Experiment*, 2023(9):093402, 2023.

[5] Giulio Biroli, Tony Bonnaire, Valentin de Bortoli, and Marc Mézard. Dynamical regimes of diffusion models. *Nature Communications*, 15:9957, 2024. doi: 10.1038/s41467-024-54281-3.

[6] Aapo Hyvärinen. Estimation of non-normalized statistical models by score matching. *Journal of Machine Learning Research*, 6:695–709, 2005.

[7] Pascal Vincent. A connection between score matching and denoising autoencoders. *Neural Computation*, 23(7):1661–1674, 2011.

[8] Herbert Robbins. An empirical Bayes approach to statistics. In *Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability, Volume 1*, pages 157–163, 1956.

[9] Koichi Miyasawa. An empirical Bayes estimator of the mean of a normal population. *Bulletin of the International Statistical Institute*, 38(4):181–188, 1961.

[10] Bradley Efron. Tweedie's formula and selection bias. *Journal of the American Statistical Association*, 106(496):1602–1614, 2011.

[11] Beatrice Achilli, Marco Benedetti, Giulio Biroli, and Marc Mézard. Theory of speciation transitions in diffusion models with general class structure, 2026.

[12] Tony Bonnaire, Raphaël Urfin, Giulio Biroli, and Marc Mézard. Why diffusion models don't memorize: the role of implicit dynamical regularization in training. In *Advances in Neural Information Processing Systems*, 2025. Oral presentation; arXiv:2505.17638.

[13] Alessandro Favero, Antonio Sclocchi, and Matthieu Wyart. Bigger isn't always memorizing: early stopping overparameterized diffusion models, 2025.

[14] Zahra Kadkhodaie, Florentin Guth, Eero P. Simoncelli, and Stéphane Mallat. Generalization in diffusion models arises from geometry-adaptive harmonic representations. In *International Conference on Learning Representations*, 2024. Outstanding Paper award.

[15] Beatrice Achilli, Enrico Ventura, Gianluigi Silvestri, Bao Pham, Gabriel Raya, Dmitry Krotov, Carlo Lucibello, and Luca Ambrogioni. Losing dimensions: geometric memorization in generative diffusion, 2024.

[16] Hugo Cui, Cengiz Pehlevan, and Yue M. Lu. A solvable model of learning generative diffusion: theory and insights, 2025.

[17] Gabriel Raya and Luca Ambrogioni. Spontaneous symmetry breaking in generative diffusion models. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

[18] Davide Ghio, Yatin Dandi, Florent Krzakala, and Lenka Zdeborová. Sampling with flows, diffusion, and autoregressive neural networks from a spin-glass perspective. *Proceedings of the National Academy of Sciences*, 121(27):e2311810121, 2024.

[19] Jérôme Garnier-Brun, Luca Biggio, Davide Beltrame, Marc Mézard, and Luca Saglietti. Biased generalization in diffusion models, 2026. Spotlight, International Conference on Machine Learning.

[20] Antonio Sclocchi, Alessandro Favero, and Matthieu Wyart. A phase transition in diffusion models reveals the hierarchical nature of data. *Proceedings of the National Academy of Sciences*, 122(1):e2408799121, 2025.

[21] Antonio Sclocchi, Alessandro Favero, Noam Itzhak Levi, and Matthieu Wyart. Probing the latent hierarchical structure of data via diffusion models. In *International Conference on Learning Representations*, 2025. arXiv:2410.13770.

[22] Alessandro Favero, Antonio Sclocchi, Francesco Cagnetta, Pascal Frossard, and Matthieu Wyart. How compositional generalization and creativity improve as diffusion models are trained. In *International Conference on Machine Learning*, 2025.

[23] Song Mei. U-Nets as belief propagation: efficient classification, denoising, and diffusion in generative hierarchical models, 2024.

[24] Jérôme Garnier-Brun, Marc Mézard, Emanuele Moscato, and Luca Saglietti. How transformers learn structured data: insights from hierarchical filtering, 2024.

[25] Judea Pearl. *Probabilistic Reasoning in Intelligent Systems*. Morgan Kaufmann, 1988.

[26] Frank R. Kschischang, Brendan J. Frey, and Hans-Andrea Loeliger. Factor graphs and the sum-product algorithm. *IEEE Transactions on Information Theory*, 47(2):498–519, 2001.

[27] Marc Mézard and Andrea Montanari. *Information, Physics, and Computation*. Oxford University Press, 2009.

[28] Martin J. Wainwright and Michael I. Jordan. Graphical models, exponential families, and variational inference. *Foundations and Trends in Machine Learning*, 1(1–2):1–305, 2008.

[29] Leonard E. Baum, Ted Petrie, George Soules, and Norman Weiss. A maximization technique occurring in the statistical analysis of probabilistic functions of Markov chains. *The Annals of Mathematical Statistics*, 41(1):164–171, 1970.

[30] Lawrence R. Rabiner. A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2):257–286, 1989.

[31] Rudolph E. Kalman. A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1):35–45, 1960.

[32] Herbert E. Rauch, F. Tung, and Charlotte T. Striebel. Maximum likelihood estimates of linear dynamic systems. *AIAA Journal*, 3(8):1445–1450, 1965.

[33] Arthur P. Dempster, Nan M. Laird, and Donald B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society, Series B*, 39(1):1–38, 1977.

[34] C. F. Jeff Wu. On the convergence properties of the EM algorithm. *The Annals of Statistics*, 11 (1):95–103, 1983.

[35] Xiao-Li Meng and Donald B. Rubin. Maximum likelihood estimation via the ECM algorithm: a general framework. *Biometrika*, 80(2):267–278, 1993.

[36] Ronald A. Fisher. Theory of statistical estimation. *Mathematical Proceedings of the Cambridge Philosophical Society*, 22(5):700–725, 1925.

[37] Thomas A. Louis. Finding the observed information matrix when using the EM algorithm. *Journal of the Royal Statistical Society, Series B*, 44(2):226–233, 1982.

[38] Olivier Cappé, Eric Moulines, and Tobias Rydén. *Inference in Hidden Markov Models*. Springer, 2005.

[39] Henry Teicher. Identifiability of finite mixtures. *The Annals of Mathematical Statistics*, 34(4): 1265–1269, 1963.

[40] Sidney J. Yakowitz and John D. Spragins. On the identifiability of finite mixtures. *The Annals of Mathematical Statistics*, 39(1):209–214, 1968.

[41] Brian D. O. Anderson. Reverse-time diffusion equation models. *Stochastic Processes and their Applications*, 12(3):313–326, 1982.

[42] Ulrich G. Haussmann and Etienne Pardoux. Time reversal of diffusions. *The Annals of Probability*, 14(4):1188–1205, 1986.

[43] Erich L. Lehmann and George Casella. *Theory of Point Estimation*. Springer, 2nd edition, 1998.

[44] Steven M. Kay. *Fundamentals of Statistical Signal Processing: Estimation Theory*. Prentice Hall, 1993.

A Notation and diffusion conventions

A.1 Symbols

symbol	meaning	symbol	meaning
L	chain length = ambient dimension	$K,\ K_\theta$	transition kernel
M	number of observed sequences	φ	innovation density
N_{g}	number of quadrature grid points	ρ	autoregressive coefficient
$u_1, \ldots, u_{N_{\mathrm{g}}}$	grid points	q	innovation variance
h	grid spacing	C	mixture components
A	grid half-width, domain $[-A, A]$	μ	initial law $\mu(a_1)$
t	diffusion time	ℓ_t	unary likelihood factor
$\Delta_t = 1 - e^{-2t}$	noise variance	$\overrightarrow{m},\ \overleftarrow{m}$	forward, backward message
$P_0,\ P_t$	clean, noised distribution	θ	kernel parameters
$S(x,t)$	score $\nabla_x \log P_t(x)$	$\mathcal{Q}$	EM auxiliary function
$\langle a \rangle_{x,t}$	posterior mean of a given x	$\boldsymbol{\Xi}$	expected transition statistic
η	standardised Gaussian white noise	$\mathcal{L}$	marginal log-likelihood
$t_{\min},\ t_{\max}$	reverse-integration endpoints	$\widehat{\mathbf{a}}_{\mathrm{LMMSE}}$	LMMSE estimator

Two symbols carry a history worth stating, because an earlier draft of this work used them differently. K **always means the transition kernel** and never a grid size; the grid size is always N_{g}. The ambient dimension is written L because it *is* the chain length and every graphical-model statement is indexed by sites; the diffusion literature more often writes n for the sample count, which here is M.

A.2 Time orientation and the reverse SDE

Convention, used without variation. η is standardised Gaussian white noise, $\langle \eta(t)\eta(t')\rangle = \delta(t-t')$, so every $\sqrt{2}$ is written explicitly.
Forward, t running from 0 to $t_{\max}$:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -x + \sqrt{2}\,\eta(t), \qquad x(t) = e^{-t}a + \sqrt{\Delta_t}\,z, \qquad \Delta_t = 1 - e^{-2t}, \quad z \sim \mathcal{N}(0, I)\,.$$

Reverse, in the elapsed-time variable $\tau = t_{\max} - t$ running from 0 to $t_{\max} - t_{\min}$, so that increasing τ means decreasing t:

$$\frac{\mathrm{d}x}{\mathrm{d}\tau} = x + 2\,S(x, t_{\max} - \tau) + \sqrt{2}\,\eta(\tau).$$

Probability-flow ODE, same marginals, deterministic: $\mathrm{d}x/\mathrm{d}\tau = x + S(x, t_{\max} - \tau)$.
Initialisation $x(\tau{=}0) \sim \mathcal{N}(0, I)$; termination at $\tau = t_{\max} - t_{\min}$.

The reverse drift follows from the time-reversal theorem for diffusions [41, 42]: for a forward SDE $\mathrm{d}x = f\,\mathrm{d}t + g\,\mathrm{d}W$ the time-reversed process has drift $f - g^2\nabla \log p_t$ and the same diffusion coefficient. With $f = -x$ and $g = \sqrt{2}$ the backward-in-t drift is $-x - 2S$, and rewriting in τ flips the sign to give the box above. Because unit-variance data keep $\mathrm{Var}(x_t) = 1$, initialising at $\mathcal{N}(0, I)$ incurs a bias $e^{-2t_{\max}}$, which is 2.5×10^{-3} at $t_{\max} = 3$. The implementation is `src/reverse.py`, whose module docstring states the same convention; the sampler uses a geometric time grid, denser at small t, because the reverse drift stiffens like $1/\Delta_t$ there.

B The score–posterior identity

Differentiation under the integral. $P_t(x) = \int \mathrm{d}a\, P_0(a)\, g_{\Delta_t}(x - e^{-t}a)$ with $g_{\Delta_t}(u) = (2\pi\Delta_t)^{-L/2} e^{-\|u\|^2/2\Delta_t}$. Since $\nabla_x g_{\Delta_t}(x - e^{-t}a) = -\frac{x - e^{-t}a}{\Delta_t} g_{\Delta_t}(x - e^{-t}a)$ and, for x in a compact neighbourhood, $\|x - e^{-t}a\|\, g_{\Delta_t}(x - e^{-t}a) \le c_1 e^{-c_2\|a\|^2}$ uniformly, the integrand is dominated by an integrable function whenever P_0 has a finite first moment. Dominated convergence then licenses

$$\nabla_x P_t(x) = \int \mathrm{d}a\ P_0(a)\, \frac{e^{-t}a - x}{\Delta_t}\, g_{\Delta_t}(x - e^{-t}a).$$

Dividing by $P_t(x)$ and recognising $P_0(a)g_{\Delta_t}(x - e^{-t}a)/P_t(x) = P(a \mid x, t)$ gives (3). Nothing is approximated. In the empirical-Bayes literature this is Tweedie's formula [8–10].

Risk decomposition. For any u that is a function of x alone,

$$\mathbb{E}\big[\|a - u\|^2 \,\big|\, x\big] = \mathbb{E}\big[\|a - \langle a\rangle_{x,t}\|^2 \,\big|\, x\big] + \|u - \langle a\rangle_{x,t}\|^2, \tag{14}$$

by expanding $a - u = (a - \langle a\rangle_{x,t}) + (\langle a\rangle_{x,t} - u)$ and noting the cross term vanishes because $\langle a\rangle_{x,t} - u$ is x-measurable and $\mathbb{E}[a - \langle a\rangle_{x,t} \mid x] = 0$. Hence $\langle a\rangle_{x,t}$ minimises the conditional risk, with an irreducible floor given by the first term [43]. Optimality is with respect to squared error specifically; a different loss selects a different functional of the posterior. This is why every table in §8 reports error *against* $\langle a\rangle_{x,t}$ rather than raw risk: the floor is common to all estimators and subtracting it is what makes the comparison informative.

C What "exact" means here, in four parts

The body compresses this to two sentences; the distinctions matter enough to be kept apart somewhere, because collapsing any two of them is how a discretised number acquires the authority of an exact one.

(i) Sum-product on (4) is *exact at the level of functional messages*, as a consequence of the graph being a tree.

(ii) A message is a *function on* $\mathbb{R}$, and this is forced by the state space, not chosen. On a discrete graphical model a_i takes finitely many values and $\overrightarrow{m}_i$ is a vector of that length. Here a_i ranges over $\mathbb{R}$, so $\overrightarrow{m}_i$ assigns a number to every real a_i: it is an unnormalised density, the evidence from everything left of site i. Only when the transition and the likelihood are both Gaussian does that function stay inside a finite-dimensional family — two numbers, a mean and a variance, which is the Kalman filter. For a general φ it does not, and the integral in (5) produces a new shape at every site. An implementation must therefore represent a function. We use a truncated uniform grid $[-A, A]$ with N_{g} points, spacing h and composite trapezoidal weights, which defines a *finite-state model* for which the recursion is *exact up to floating-point arithmetic*.

(iii) That finite-state model *approximates* the continuous one, with an error that must be measured — §4 measures it.

(iv) Monte Carlo enters this note in exactly one place: when a statistic is computed from *generated samples*. Every other number here — posterior means, evidences, information matrices — comes from deterministic quadrature and carries no sampling error at all, only discretisation error. Where a quantity is averaged over random draws of a dataset, that is replicate spread and is labelled as such.

We write "exact tree inference" for (i), "exact for the discretised model" for (ii), and "grid-BP reference" for the numerical arm; none is abbreviated to "exact".

D Sum-product on the posterior chain

Posterior factorisation. Bayes' rule applied to (2), with the likelihood factorising sitewise because the forward noise is independent across coordinates, gives (4). Each likelihood factor has degree one among latent variables, so attaching it to the prior's path adds a leaf and cannot create a cycle: the posterior factor graph is a tree. Conditioning usually creates dependence; it does not here, precisely because each observation is generated from a single latent variable. The construction fails the moment the forward noise is correlated across coordinates, since then $P(x \mid a)$ does not factorise, the likelihood becomes a factor touching several a_i, and the posterior graph acquires edges.

Exactness on a tree. Let the graph be a tree and let a_i be a variable node. Removing a_i disconnects it into components, and every factor involves a_i together with variables of exactly one component — a factor touching two components would create a second path between them, hence a cycle. Grouping factors by component gives $P \propto \prod_c g_c(a_i, v^{(c)})$, so conditionally on a_i the components

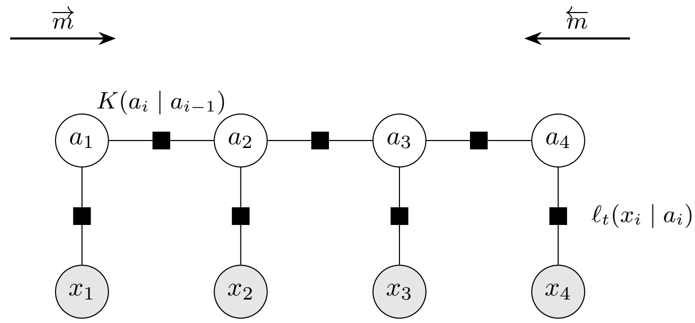

Figure 3: Factor graph of the posterior (4). Open circles are latent variables, shaded circles are observations, squares are factors. Each transition factor $K(a_i \mid a_{i-1})$ joins two adjacent latent variables; each observation enters through a *unary* factor ℓ_t attached to a single latent variable. Conditioning therefore reweights node potentials without creating edges among the a_i, and the posterior graph remains a chain. Arrows show the two message sweeps (5).

are independent. Unrolling the forward recursion (5),

$$\overrightarrow{m}_i(a_i) \propto \int \mathrm{d}a_{1:i-1}\ \mu(a_1) \prod_{j=2}^{i} K(a_j \mid a_{j-1}) \prod_{j=1}^{i-1} \ell_t(x_j \mid a_j),$$

which is the joint density of everything at or left of site i with the left variables integrated out and $\ell_t(x_i \mid a_i)$ *not* included; symmetrically for $\overleftarrow{m}_i$. The two are conditionally independent given a_i, so their product times the local evidence is proportional to the marginal. The pairwise statement is the same argument with the separator taken to be the edge.

Where the local factor goes. In (5) the local evidence at site $i-1$ is absorbed *before* the transition, so $\overrightarrow{m}_i$ carries the factors $\ell_t(x_1 \mid a_1), \ldots, \ell_t(x_{i-1} \mid a_{i-1})$ and $\overleftarrow{m}_i$ carries $\ell_t(x_{i+1} \mid a_{i+1}), \ldots, \ell_t(x_L \mid a_L)$, and each appears exactly once in the marginal. Absorbing it *after* would place $\ell_t(x_i \mid a_i)$ in both messages and square it: the belief would be sharper than the truth and every posterior variance too small. This produces plausible-looking output, which is why the convention is pinned by `tests/test_message_normalization.py` rather than left to a comment.

Evidence. Messages are renormalised at each step to prevent underflow; accumulating the discarded constants recovers $\log P_t(x)$ for the discretised model exactly. This is the objective that §5 maximises, and having it in closed form is what makes it possible to check a gradient against a finite difference of the thing being optimised. `src/bp_grid.py` $\to$ `grid_bp` returns it alongside the beliefs; `src/exact_scores.py` $\to$ `gaussian_log_evidence` is the closed form it is tested against.

Complexity at the functional level. Each message update integrates one variable against the kernel. Abstractly this is an integral operator applied to a function; represented on N_g points it is one $N_\mathrm{g} \times N_\mathrm{g}$ matrix–vector product, $O(N_\mathrm{g}^2)$. There are $2L$ updates, so one sequence costs $O(LN_\mathrm{g}^2)$ against $O(N_\mathrm{g}^L)$ for naive marginalisation. At $L = 32, N_\mathrm{g} = 401$ that is about 5×10^6 operations. Batching over sequences turns each update into a single GEMM, which is why the same code runs on a GPU without restructuring (`src/backend.py`, gated by `tests/test_backend_parity.py`).

E Covariance-stationary versus strictly stationary

Two notions of stationarity coincide for Gaussian processes and separate here, and the body relies on the weaker one, so the difference is worth stating.

A process is *strictly stationary* if the joint law of $(a_{i+1}, \ldots, a_{i+k})$ does not depend on i, for every k. It is *covariance-stationary* (or weakly, or second-order stationary) if only the first two moments are shift-invariant: $\mathbb{E}[a_i]$ constant and $\mathrm{Cov}(a_i, a_j)$ a function of $|i-j|$ alone. Strict stationarity implies the covariance version whenever second moments exist; the converse is false in general.

Why they coincide in the Gaussian case. A Gaussian law is determined by its mean vector and covariance matrix. If the mean is constant and the covariance depends only on $|i-j|$, then the joint law of any window $(a_{i+1}, \dots, a_{i+k})$ is the Gaussian with that mean and that covariance block — and the block depends on i only through the differences $|j-j'|$, which are shift-invariant. So the joint law is shift-invariant, which is strict stationarity. The implication runs through "determined by two moments" and therefore fails for any family that is not.

What holds for our chain. With $q = 1-\rho^2$ and $\mathrm{Var}(a_1) = 1$, second moments propagate exactly:

$$\mathrm{Var}(a_i) = \rho^2 \,\mathrm{Var}(a_{i-1}) + q = \rho^2 + (1-\rho^2) = 1,$$

by induction, and $\mathrm{Cov}(a_i, a_j) = \rho^{|i-j|}$, *whatever the shape of* φ. The chain is therefore covariance-stationary, and this is exactly the property the variance-matched family needs: Gaussian, Laplace, Student, uniform and bimodal innovations share the same second-order structure and differ only beyond it.

It is **not** strictly stationary for non-Gaussian φ. The invariant law of $a_i = \rho a_{i-1} + \varepsilon_i$ is that of $\sum_{k\geq 0} \rho^k \varepsilon_{-k}$, an infinite convolution which is not Gaussian, so drawing $a_1 \sim \mathcal{N}(0,1)$ starts the chain off its invariant law. The marginal of a_i then drifts towards it over a burn-in of order $1/\log(1/\rho) \approx 6$ sites at $\rho = 0.85$. This does not affect the comparisons in the body, because every family shares the same a_1 law and the same covariance and so still differs only beyond second moments — but $q = 1-\rho^2$ alone does not deliver strict stationarity, and we do not claim it does.

F The Gaussian model and the second-order baseline

For $\varphi = \mathcal{N}(0,q)$ with $q = 1-\rho^2$ and $a_1 \sim \mathcal{N}(0,1)$, the vector a is jointly Gaussian with $(\Sigma_0)_{ij} = \rho^{|i-j|}$. Expanding $P_0(a) \propto \exp[-\frac{1}{2q}\sum_{i\geq 2}(a_i - \rho a_{i-1})^2 - \frac{1}{2}a_1^2]$ produces only a_i^2 and $a_i a_{i-1}$ terms, so Σ_0^{-1} is tridiagonal with $(\Sigma_0^{-1})_{ii} = (1+\rho^2)/q$ in the interior, $1/q$ at the two ends, and $-\rho/q$ on the off-diagonals. Adding the likelihood contributes $(e^{-2t}/\Delta_t)I$, so the posterior precision is tridiagonal at every t: Markovianity is a statement about the precision, while the covariance Σ_0 is dense.

Since $x = e^{-t}a + \sqrt{\Delta_t}z$ with z independent, (a,x) is jointly Gaussian with $\mathrm{Cov}(a,x) = e^{-t}\Sigma_0$ and $\mathrm{Cov}(x) = e^{-2t}\Sigma_0 + \Delta_t I =: \Sigma_t$, whence $\mathbb{E}[a \mid x] = e^{-t}\Sigma_0\Sigma_t^{-1}x$ and, by (3), $S(x,t) = -\Sigma_t^{-1}x$. For zero-mean jointly Gaussian variables the conditional expectation is linear, so it coincides with the LMMSE estimator [44], which proves the second half of Proposition 1. The first half is the observation that both maps in (7) read only the mean and variance of φ, so the whole recursion is a function of $(0,q)$ alone and therefore returns what it returns on the Gaussian chain.

Consequently, on this model, the *message-representation error* and the *model error* coincide: projecting messages onto Gaussians and replacing the prior by its covariance-matched Gaussian give the same estimator. This is a property of single-Gaussian closure with linear transitions, not of Gaussian approximations in general, and the two separate for richer message families.

G Numerical representation

Messages are represented on a uniform grid $u_1, \dots, u_{N_\mathrm{g}}$ spanning $[-A, A]$ with spacing $h = 2A/(N_\mathrm{g}-1)$ and composite trapezoidal weights $w_k = h$ for interior points, $h/2$ at the two ends. All recursions run in the log domain with a per-step maximum subtracted before exponentiating, and the removed constants are restored in the evidence, so no message underflows.

Two error sources. *Truncation*: probability mass outside $[-A, A]$ is discarded rather than transported, an error controlled by A and essentially independent of N_g. *Quadrature*: the trapezoidal rule is second order in h for integrands with bounded second derivative, an error controlled by h at fixed A. Figure 4 measures them separately, using the worst-case boundary mass over sites for the first and the interior column-mass residual of the transition matrix for the second; the interior restriction $|u| \leq A/2$ removes columns whose transition density has a tail outside the grid, which is what makes the second diagnostic measure quadrature alone. The observed rates are: boundary

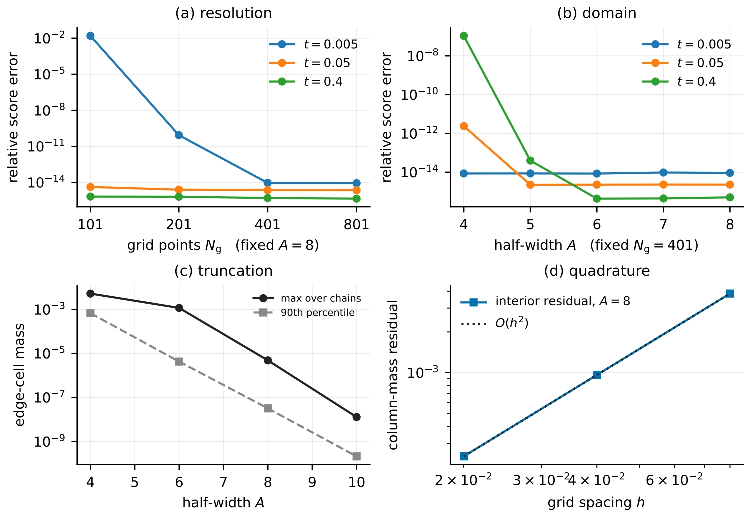

Figure 4: Grid and domain control, Laplace chain. (a) Relative score error against a high-resolution reference at fixed $A = 8$; (b) the same at fixed $N_{\mathrm{g}} = 401$ as the domain grows. Both saturate at the floating-point floor $\sim 10^{-14}$, so beyond that the curves measure arithmetic, not discretisation. (c) Worst boundary mass over sites — the truncation diagnostic — against A. (d) Interior column-mass residual against spacing at $A = 8$ — the quadrature diagnostic — with an $O(h^2)$ reference. The two diagnostics do different jobs and neither alone certifies convergence. All values are numerical, not empirical. **[PENDING: regenerate from the frozen batch]**

mass $3 \times 10^{-4} \to 1.4 \times 10^{-8} \to 1.4 \times 10^{-10}$ at $A = 4, 8, 10$; interior residual falling by a factor of four per halving of h, i.e. $O(h^2)$.

At $A = 4$ the interior residual saturates at $\sim 10^{-3}$ instead of continuing to fall, because at that domain even the "interior" columns have truncated tails — the floor is the truncation error leaking into the quadrature diagnostic, and it is visible in the data (`outputs/exp_18/boundary.csv`). This is the reason the two diagnostics have to be read together and neither alone certifies convergence.

What is not claimed. A grid-resolution sweep at fixed A does not rule out truncation error, and a domain sweep at fixed N_{g} does not rule out quadrature error. We report both axes and the two diagnostics; we do not claim continuum convergence, only that at $N_{\mathrm{g}} = 401$, $A = 8$ both diagnostics sit far below every effect the paper reports. For the Laplace family, whose innovation density has a discontinuous derivative at the origin, we claim second order and no better.

H Learning procedure

Objective. Maximum marginal likelihood $\mathcal{L}(\theta) = \sum_\mu \log P_{t,\theta}(x^\mu)$, with the complete-data objective $\log P_\theta(a, x) = \log \mu(a_1) + \sum_{i \geq 2} \log K_\theta(a_i \mid a_{i-1}) + \sum_i \log \ell_t(x_i \mid a_i)$. Only the middle term depends on θ, which is why Ξ of (9) is sufficient and why $\mathcal{Q}(\theta \mid \theta') = \langle \Xi(\theta'), \log K_\theta \rangle$ up to θ-free constants.

E-step. One forward and one backward sweep per sequence, then a rank-one accumulation of the pairwise statistics: writing $\ell^{(i)}$ for the vector $\ell_t(x_i \mid \cdot)$ on the grid, and with $f_i = w \odot \overrightarrow{m}_i \odot \ell^{(i)}$ and $g_i = w \odot \overleftarrow{m}_i \odot \ell^{(i)}$, the pairwise belief on edge $(i, i+1)$ is $f_i(j)\, K[k, j]\, g_{i+1}(k)/Z_i$ with $Z_i = g_{i+1}^\top K f_i$, and $\Xi = \left(\sum_{\mu,i} g_{i+1}(f_i)^\top / Z_i\right) \odot K$. Ξ sums to the number of edges in the dataset, which is asserted in the tests. Exact for the discretised model, in the sense of Appendix C(ii). Implementation `src/em.py` $\to$ `e_step`.

M-step, by kernel.

- *Gaussian*, $\theta = (\rho, q)$: closed-form maximiser, the belief-weighted Yule–Walker equations $\rho = S_{01}/S_{00}$, $q = (S_{11} - \rho S_{01})/W$. Exact maximisation, hence exact EM for that family.
- *Laplace*, $\theta = (\rho, b)$: closed form despite non-differentiability. Profiling b out of $\mathcal{Q}$ turns the problem into minimising the belief-weighted mean absolute residual, whose exact solution is a weighted median. Exact maximisation.
- *Mixture* (12): the complete-data problem carries a second latent variable, so the M-step is itself an inner EM over the mixture label with Ξ held fixed. We run **four** inner conditional-maximisation sweeps, alternating the (π, ν, s^2) block and the ρ block, both closed form. This increases $\mathcal{Q}$ without maximising it: **generalised EM / ECM** [35], not exact EM.
- *Mixture-density network*: one backtracked Adam step, accepted only if $\mathcal{Q}$ increased, halving the step up to five times otherwise and standing still if no ascent is found. Also generalised EM.

Guarantees, initialisation, stopping. Generalised EM guarantees monotone ascent of $\mathcal{L}$ [33, 34]; convergence is to a stationary point, not necessarily a global maximum. Mixture components are initialised spread over $[-\sqrt{\mathrm{Var}}, \sqrt{\mathrm{Var}}]$ with random scales; ρ is initialised away from its truth in every experiment — at 0.3 in most, and over $\{0, 0.3, 0.6, -0.4\}$ in the initialisation sweep of `exp_18`. The truth is 0.85 in `exp_02`, `exp_16`, `exp_18` and `exp_22`, and 0.8 in `exp_06`, `exp_07` and `exp_08`; every table and figure states the value it used. Iteration stops when the relative increase of $\mathcal{L}$ falls below 10^{-9}, capped at 50–120 iterations depending on the experiment. Variances are floored at 10^{-6} to prevent component collapse; no other regularisation is applied, and the component means are unconstrained (§5). Monotonicity is recorded per run as the largest decrease across iterations and is zero in every run reported here.

I Convergence rates, clean against channel

Two slownesses, measured separately, because they are independent and were once confused for one another.

Optimisation: coordinates of the same kernel settle at very different rates. Running EM to 800 updates at $M = 512$, $C = 8$ and recording the whole parameter trace, we measure for each coordinate the last update at which it was still further than a fixed relative tolerance from its final value. Over sixteen seeds the median is 80 updates for ρ (10^{-3} tolerance), 68 for the innovation variance (10^{-2}), and 229 for the excess kurtosis (2×10^{-2}), the last ranging up to 638. The shape is thus roughly three times slower to settle than the correlation, and monitoring ρ — the natural thing to plot — shows a converged-looking trace over a kernel that is not. This is why the stopping rule may not be read off ρ, and why an iteration budget must be justified against the *slowest* coordinate.

Statistics: fitting through the channel costs rate, not just constant. Comparing maximum likelihood on clean transition pairs against the same estimate obtained through the noising channel, at sixteen replicates over $M = 64$ to 1024, the fitted log–log slopes are

	ρ	innovation variance
clean transition pairs	-0.52	-0.50
through the channel	-0.44	-0.36

The clean arm sits on the parametric $M^{-1/2}$ to two decimal places on both coordinates, which is the check that the estimator itself is not the bottleneck; the noised arm is slower, and more so on the shape than on the correlation. Note that the clean arm is *not* flat in M — its error falls by a factor of 3.4 on ρ and 4.3 on the variance across the range — which is worth stating explicitly because an earlier three-replicate version of this comparison appeared flat and was misread as a discretisation floor.

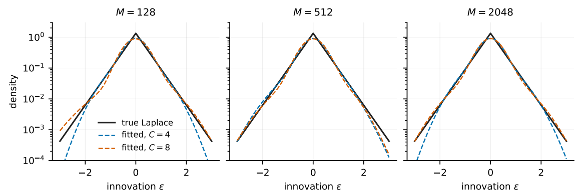

Figure 5: The fitted innovation density against the true Laplace density, on a log scale so the tails are visible. What is recovered here is a density, not a parameter: the mixture is a representation of it.

J The rotating ring

Two clocks. Internal time $u = 0, \ldots, T-1$ runs along the trajectory; diffusion time t is the noise level. One sample is a *whole* trajectory $\mathbf{a} \in \mathbb{R}^{2T}$, so the ambient dimension is $D = 2T$ and the channel hits all of it at once. This is the point of the model: the object being estimated is a property of the trajectory, not of any frame.

The gauge. Let $U_\psi = \operatorname{blockdiag}(R_{-u\psi})_{u=0}^{T-1}$, which rotates frame u backwards by $u\psi$. It is orthogonal, so it commutes with the isotropic channel: if $\mathbf{x} = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\,\xi$ then $U_\psi \mathbf{x} = e^{-t} U_\psi \mathbf{a} + \sqrt{\Delta_t}\, U_\psi \xi$ and $U_\psi \xi \overset{d}{=} \xi$. Applying it to (13) turns $z_{u+1} = R_\psi z_u + \sigma \eta_u$ into a driftless walk, so

$$P_t(\mathbf{x} \mid \psi) \;=\; P_t^0(U_\psi \mathbf{x}),$$

with P_t^0 the $\psi = 0$ density. At known ψ the rotation costs nothing statistically.

The $\psi = 0$ density in closed form. Conditionally on z_0 the trajectory is Gaussian with a z_0-independent covariance, so P_0 is a two-parameter *location* mixture of one Gaussian. Writing $X \in \mathbb{R}^{T\times 2}$ for the gauged trajectory, $A_t = e^{-2t}\sigma^2 K + \Delta_t \mathbf{I}_T$ with $K_{uv} = \min(u, v)$, $g = A_t^{-1}\mathbf{1}$, $\kappa = \mathbf{1}^\top g$ and $\beta = \|X^\top g\|$,

$$\log P_t^0(X) \;=\; -\tfrac{1}{2}\operatorname{tr}(X^\top A_t^{-1} X) + \log \mathcal{Z}_t(\beta) + \text{const}, \qquad \mathcal{Z}_t(\beta) = \int_0^\infty \rho\, e^{-(\rho-1)^2/2\lambda}\, e^{-e^{-2t}\kappa\rho^2/2}\, I_0(e^{-t}\rho\beta)\,\mathrm{d}\rho,$$

a quadratic form plus one one-dimensional Bessel integral. The constant depends on t but not on ψ, so it cancels in every responsibility. K is singular with a vanishing zeroth row — frame 0 *is* z_0 and carries no walk noise — so A_t is exactly block-diagonal and frame 0 decouples at every t.

Proof of Theorem 1. Unrolling (13),

$$z_u \;=\; R_\psi^u z_0 \;+\; \sigma \sum_{k=1}^{u} R_\psi^{u-k} \eta_k .$$

Each η_k is isotropic Gaussian, and a rotation of an isotropic Gaussian has the same law, so $R_\psi^{u-k}\eta_k \overset{d}{=} \eta_k$, independently across k. And θ_0 is uniform on $[0, 2\pi)$, so z_0 has a rotationally invariant law and $R_\psi^u z_0 \overset{d}{=} z_0$. The two terms are independent, so

$$z_u \;\overset{d}{=}\; z_0 + \sigma\sqrt{u}\,\xi, \qquad \xi \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_2)\,,$$

in which ψ does not appear. The channel acts coordinatewise and independently of ψ, so the noised frame satisfies $x_u \overset{d}{=} e^{-t}(z_0 + \sigma\sqrt{u}\,\xi) + \sqrt{\Delta_t}\,\xi'$, again free of ψ. Hence for every u and every t the marginal law of x_u is the same function of the model's other parameters regardless of ψ, so $\partial_\psi \log P_t(x_u) \equiv 0$ and the Fisher information about ψ carried by any single frame is exactly zero. Summing per-frame log-densities cannot create information that no term contains, so a model built from per-frame marginals has zero information about ψ. $\square$

What the theorem does not say. It does not say the marginals are uninformative in general — they determine λ and σ perfectly well, and the per-frame law is a visible non-Gaussian ring whose radius grows as $\sqrt{u}$. It says only that they are blind to the *rotation*. Nor does it say the joint is easy: the joint carries the information, and extracting it is what the estimator does.

EM over $p(\psi)$. With ψ discretised on a uniform grid of $|\psi|$ angles on $[-\pi, \pi)$, the algorithm is textbook EM for a finite mixture whose components are indexed by the rotation:

$$\text{E:} \quad r_\mu(\psi_k) = \frac{p(\psi_k)\, P_t(\mathbf{x}^\mu \mid \psi_k)}{\sum_l p(\psi_l) P_t(\mathbf{x}^\mu \mid \psi_l)}, \qquad \text{M:} \quad p(\psi_k) \leftarrow \frac{1}{M} \sum_\mu r_\mu(\psi_k).$$

Since $P_t(\mathbf{x}^\mu \mid \psi_k)$ does not depend on p, the $M \times |\psi|$ matrix of log-densities is formed once and every iteration is a reweighting — the same sufficiency that makes $\boldsymbol{\Xi}$ a one-shot statistic on the chain. Initialisation is uniform, which encodes no knowledge of where the species are and is also the fixed point of the blind arm, so that arm visibly stands still. Implementation `src/ring.py` $\rightarrow$ `log_pt_conditional` and `experiments/exp_28_ring_em.py`; the port is checked bit-for-bit against the independently audited implementation in `research/board-3problems`.

K Experimental protocol

Every number here comes from one frozen configuration, defined in a single file and imported by every experiment so that no run can silently diverge from it: Laplace chain unless stated, $\rho = 0.85$, $L = 32$, $N_\text{g} = 401$, $A = 8$, twelve log-spaced noise levels on $[0.05, 3.0]$, sixteen replicates, budgets $M \in \{32, \ldots, 4096\}$, and 256 held-out sequences on a disjoint seed. The ring uses $T = 12$, so $L = 24$. Architectures, optimiser and seeds are in Appendix K.

The stopping rule, stated once. EM stops when the fitted innovation excess kurtosis stops moving, $|\Delta\kappa_4| < 10^{-3}$, with a relative log-likelihood tolerance of 10^{-9} as a secondary guard and a cap of 400 iterations. Stopping on ρ is not permitted anywhere: by Remark 1 it certifies a converged-looking trace over an unconverged kernel. The rule is the same at every rung.

chain	Laplace AR(1), $\rho = 0.85$, $q = 1 - \rho^2$, $a_1 \sim \mathcal{N}(0,1)$, $L = 32$
other families	Gaussian, Student-t ($\nu = 5$), uniform, symmetric two-component mixture ($\kappa = 0.6$) all variance-matched to q, so $\mathrm{Cov}(a_i, a_j) = \rho^{\lvert i-j \rvert}$ exactly
grid	$N_\text{g} = 401$, $A = 8$ ($h = 0.04$); refinements $N_\text{g} \in \{801, 1601\}$
noise levels	$t \in \{0.1, 0.2, 0.4, 0.8, 1.6\}$ for training and pointwise evaluation
budgets	$M \in \{32, 128, 512, 1024, 2048, 4096\}$
test set	256 held-out sequences, seed `exp07-test`, disjoint from all training seeds
replicates	six; the replicate index is mixed into every training seed, so replicates vary the training set *and* the initialisation; the test set is common to all six
EM–BP	mixture-innovation kernel (12), $C = 4$ unless swept; 12 free parameters ρ initialised at 0.3; 50–120 EM iterations; tolerance 10^{-9}
global MLP	fully connected, 25,248 parameters, denoising score matching, 20,000 steps both parameterisations (ε and a) trained; Adam
local CNN	weight-shared one-dimensional convolution, radius r, receptive field $2r + 1$
capacity control	24 configurations, 3.2k–297k parameters
hardware	CPU: single-threaded, `medium_cpu` partition, Bocconi HPC GPU: CUDA 12.4, CuPy 13.6, gated by a CPU/GPU parity test before any sweep

Selection rules, stated plainly. There is **no validation split**. Two selections therefore happen on the evaluation set and both favour the baselines: (i) the sample-efficiency comparison reports, at each noise level, the better of the two denoising-score-matching parameterisations, which is oracle post-selection over two models; (ii) the convolution's receptive field is a fixed configuration value read off the same sweep that is evaluated on the test set. The numbers in §8 that depend on (ii) are labelled exploratory. Replacing both with validation-based selection is the first item of the revision plan.

L Scope conditions

The initial law is held fixed at the one the generating process uses, so nothing is misspecified there and nothing shows it could be recovered. The linear form of (12) is assumed: what is free is a scalar and a density, not an arbitrary bivariate transition. The chain is covariance-stationary rather than strictly stationary. Identifiability is a theorem at Rung 1, conditional at Rung 2, and assumed at Rung 3. The implementation is a discretisation whose error, for the Laplace family, is second order rather than spectral. Inference costs $O(LN_{\mathrm{g}}^2)$ per sequence, far above a network forward pass, so the estimator's advantage is in data and not in wall-clock. And the likelihood is a *composite* objective — each clean chain observed at several noise levels, the groups treated as independent — so every marginal is correctly specified but the effective sample size is the number of clean chains, not that times the number of levels.

M Reproducibility map

claim	script (`experiments/`)	output (`outputs/`)
grid and domain convergence	`exp_01_grid_validation`	`exp_01_.../grid_heatmap.csv`
boundary and quadrature diagnostics	`exp_18_... --parts boundary`	`exp_18/boundary.csv`
closure error vs t	`exp_02_laplace_gaussian_...`	`exp_02_.../laplace_summary.csv`
closure error by family	`exp_03_nongaussian_...`	`exp_03_.../innovation_sweep.csv`
EM monotonicity, ρ recovery	`exp_18_... --parts emtrace`	`exp_18/em_trace.csv`
learned innovation density	`exp_18_... --parts density`	`exp_18/innovation_density.csv`
parameter-recovery rate	`exp_06_em_parameter_...`	`exp_06_.../em_rate.csv`
Fisher information vs t	`exp_06_em_parameter_...`	`exp_06_.../price_of_noising.csv`
Fisher identity vs finite diff.	`exp_08_gradient_vs_...`	`exp_08_.../gradient_vs_exact.csv`
sample efficiency (§8)	`exp_07_em_vs_score_...`	`replicates/merged_summary.csv`
network capacity control	`exp_07_em_vs_score_...`	`exp_07_.../capacity.csv`
inference and training cost	`exp_07_em_vs_score_...`	`exp_07_.../` `{inference,training}_cost.csv`
receptive field (exploratory)	`exp_12_receptive_field`	`exp_12_.../efficiency_three_way.csv`
sampler step convergence	`exp_16_... MODE=calibrate`	`exp_16/calibrate_steps*/steps.csv`
generation by budget	`exp_16_... MODE=generate`	`exp_16/gen_n*_seed*/generation.csv`
generation by family	`exp_16_sampling_validation`	`exp_16/family_*/generation.csv`
capacity: generation	`exp_16_sampling_validation`	`exp_16/components_C*/generation.csv`
capacity: pointwise	`exp_16_sampling_validation`	`exp_16/cpoint_C*/pointwise.csv`
discrete-alphabet variant	`exp_10_discrete_alphabet`	`exp_10_.../discrete_*.csv`

Every figure in this note is produced by `paper/figures/make_figures.py`, which reads the outputs above and fails loudly if one is missing; no number in any figure is entered by hand. Exact commands, environment and HPC job configuration are in `REPRODUCIBILITY.md`, and the claim-level audit behind this revision is `REVISION_AUDIT.md`.